



Camera Technology
Focused on Scientific Imaging
and Challenging Inspection

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Dhyana 2100

User Manual

V1.0.0



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1. INTRODUCTION

1.1. Disclaimer

To protect the legitimate rights and interests of users, please carefully read our accompanying instructions, disclaimers, and safety instructions before using our company's products. This camera user manual document contains basic information about the camera, installation instructions, product features, and maintenance, aiming to make it more convenient for users to use the TUCSEN's camera. This document is only disclosed for the above purposes. Please make sure to follow the instructions and safety instructions when operating this product.

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1.2. Safety and Warning Information

Operation and Use



Caution

- Do not drop, disassemble, repair or replace internal components on your own. Such action may damage the camera components or cause personal injury.
- In the event of spillage on the equipment, please disconnect the equipment and immediately contact the nearest dealer or manufacturer for technical assistance.
- Do not touch the device with wet hands, as it may cause electric shock.
- Do not let children touch the device without supervision.
- Ensure that the temperature of the camera is within the specified temperature range for use to avoid damage.

Installation and maintenance



Caution

- Please do not install it in dusty and dirty areas or near air conditioning or heaters to reduce the risk of camera damage.
- Avoid installation and operation in extreme environments such as vibration, high temperature, humidity, dust, strong magnetic fields, explosive/corrosive gases or gases.
- Do not apply excessive vibration and impact to the equipment. This may damage the equipment.
- Do not install equipment under unstable lighting conditions. Severe lighting changes can affect the quality of the images generated by the device, avoiding high-energy lasers directly hitting the camera sensor.
- Do not use solvents or diluents to clean the surface of the equipment, as this will damage the surface of the casing.
- Please ensure that there is at least 10 cm of space around the device ventilation opening to ensure airflow flow. Do not block the ventilation openings of the equipment during use, otherwise it may cause internal temperature to be too high and damage the equipment.

Power



Caution

- Please use the original power adapter of the camera, as using an mismatched power source may damage the camera.
- If the voltage applied to the camera is greater than or less than the nominal voltage of the camera, the camera may be damaged or malfunction.
- Please refer to the specification table for the nominal voltage of the camera.

2. Product Specifications

2.1. Packing List

Item Name	Specification	Quantity	Image
sCMOS scientific grade camera	Dhyana 2100	1	
Adapter Ring	M58 adapter	1	
Power Adapter	24V/6.67A	1	
Power cord	250V/10A, 1.8m	1	
USB flash drive	Software and driver included	1	

CoaXPress frame grabber (optional)	Euresys	1	
CoaXPress Cable	3 meters, 4 for one card	4	
External Trigger Cable	HR10A-7P-6S, 6 pole plug with pins single header cable	1	

2.2. Quantum Efficiency Curve

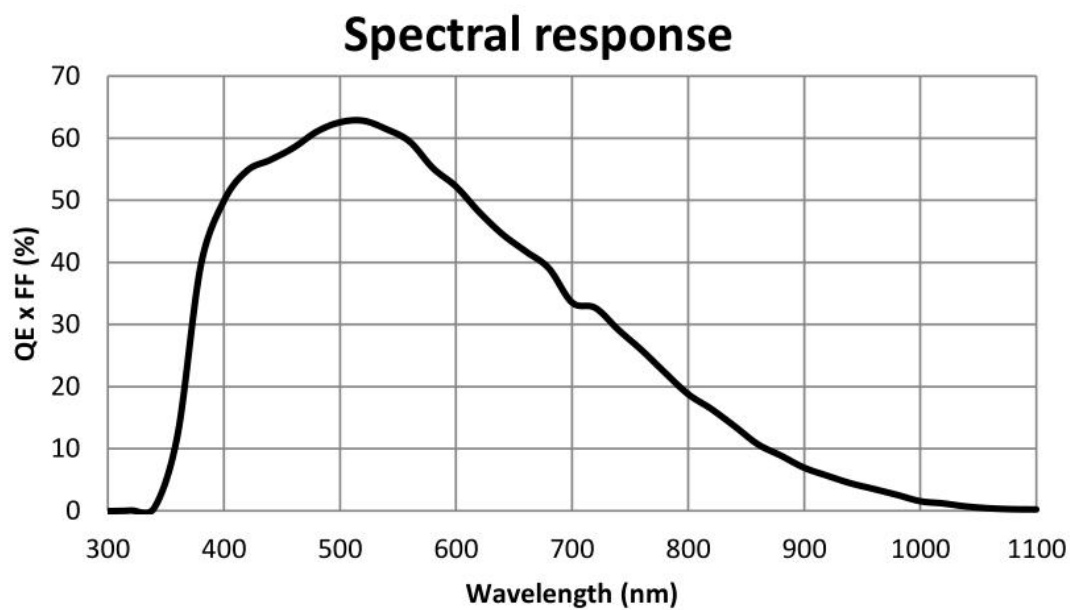


Figure -21 Dhyana 2100 Camera Quantum Efficiency Curve

2.3. Camera power and signal connections



Fig. 2-4 Camera Interface Diagram

Serial number	Name	Function
1	DC 24V	Power supply interface
2	TRIGGER	Trigger Interface
3	CXP12 x4/x8	Camera Data Transfer
4	Indicator light	Indicates if the imager is functioning properly

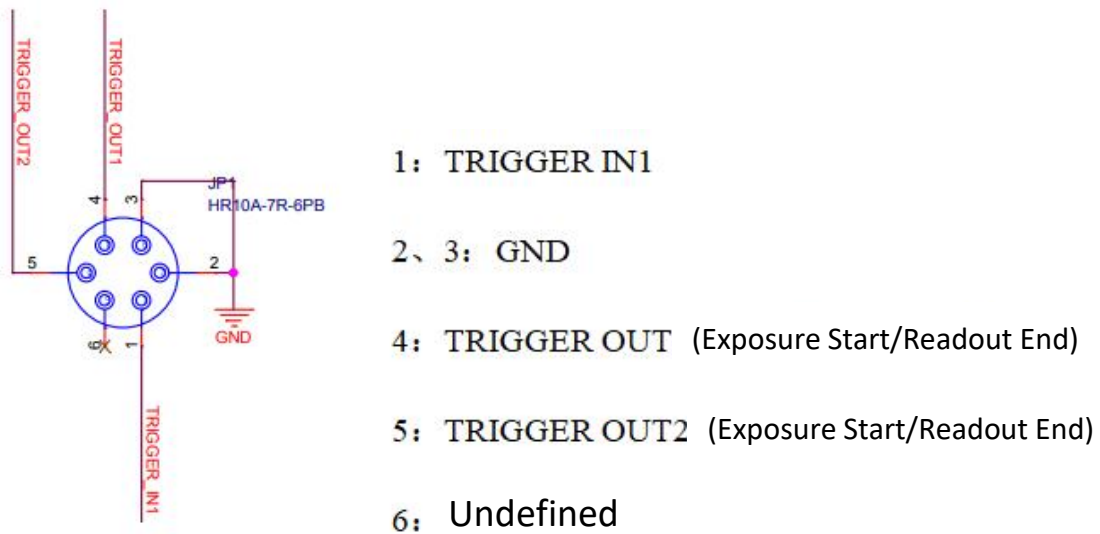


Figure2 -5 Camera Trigger Interface Definition Diagram

Note: The water-cooling ports do not distinguish between inlet and outlet ports:

1. The water-cooling port does not distinguish between the water inlet and outlet, and the water-cooling pipe is installed when it is inserted twice with a sense of stuttering.
2. With a single frame grabber, four wires must be connected to the "host" device use the camera. A single wire connected to the "follower" cannot be used.
3. For a dual frame grabber with 8 lines, lines ABCD correspond to the "host" and "follower" ① ② ③ ④.

3. Features and Functions

3.1. Camera Introduction

The Dhyana 2100 is the ideal combination of high speed and high resolution. Not only does it reach a speed of 450 fps at 21 million full resolution, but it also features a high-speed

binning mode especially designed for low-light applications, with higher sensitivity and dynamic range, and a speed of up to 975 fps. It is suitable for imaging applications such as semiconductor/wafer inspection, high-speed aerial photography, and flat panel detection (FPD).

3.2. Camera Main Features

- High-speed high resolution: 450 fps @ 21 MP with the ability to view sample details at high motion speeds
- High speed binning mode: up to 975 fps
- High sensitivity: 63% @ 520nm
- Global shutter mode: high speed image capture without artefacts and distortion
- High dynamic range: 68.8 dB
- Deep TEC cooling to suppress dark noise: over 30% cooling depth (°C)
- Advanced air-cooled & water-cooled heat dissipation methods: reduce dark current noise, minimize vibration, and contribute to the stable operation of the instrument system.
- CXP12 x4/x8 high speed interface
- Compliant with GenICam™ standard
- Comply with CoaXPress standard, supporting the mainstream acquisition card on the market.
- Support ROI
- Support Binning
- Flat field correction
- Dark field correction (Sensor)
- Supports test image output
- Camera temperature monitoring

3.3. Defective Pixel Correction (DPC)

There are always a few anomalous values on the sCMOS camera sensor, which can be corrected by the camera's Defective Pixel Correction (DPC) function. When turned on, DPC corrects anomalous values in the image. However, it may cause pixel flicker in some single-molecule imaging applications for which the DPC function is not recommended, or only the weakest grade of correction is used.

The Dhyana 2100 utilizes a static Defective Pixel Correction function by correcting the matrix pixels and is currently open for Low stops.

3.4. Dark Signal Non-Uniformity (DSNU)

When the camera acquires an image in complete darkness, in an ideal image, all pixel gray values should be close to zero and should be equal. However, in practice when the camera acquires images in the dark, subtle differences in the performance of each image element in the sensor will result in some variation in the gray value of the pixels output from the camera.

In practice, when there are no photons incident on the camera, the images obtained usually do not show a 0 gray value (DN). This is because the manufacturer usually sets a background value for the camera, e.g., 100 gray values, to which the effect of noise on the measurement is added or subtracted when there is no light. However, without careful calibration and correction, this fixed bias may vary somewhat from pixel to pixel. This variation is called "fixed graphical noise" and can be measured as dark signal non-uniformity (DSNU). It represents the standard deviation of the pixel bias and is measured in units of charge.

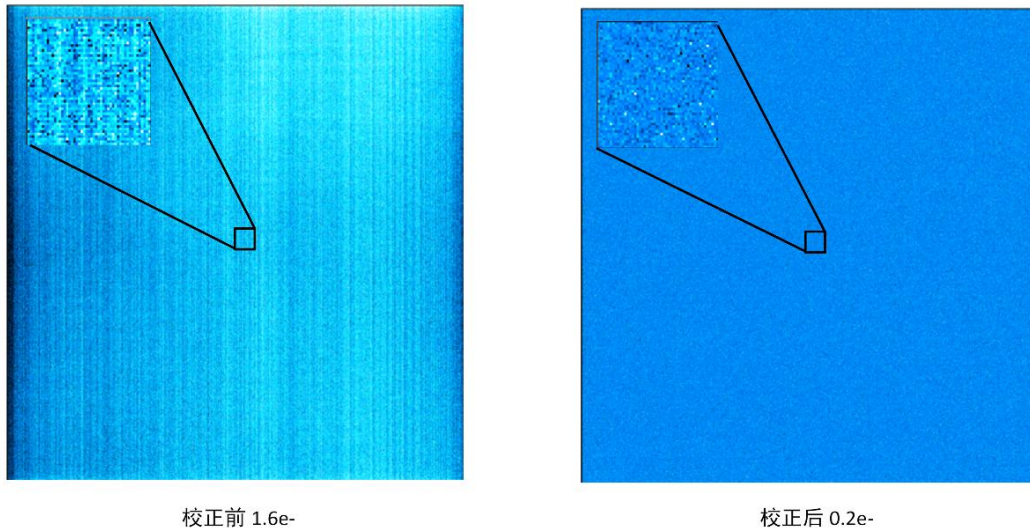


Figure3 -1 Comparison before (left) and after (right) DSNU correction.

For many low-light imaging cameras, the DSNU is typically below 0.5 e-. This means that in medium or high light level applications (where each pixel is typically capable of capturing hundreds or thousands of photons) then the effect of this noise is completely negligible. And even for low light applications, if the DSNU is lower than the camera's readout noise (typically 1-3 e-), then it is unlikely that this fixed graphical noise will have an impact on image quality.

3.5. Photo-Response Non-Uniformity (PRNU)

When the camera photographs a uniformly light-colored target in bright light, ideally the image should have all pixel gray values close to and equal to the maximum gray value. In practice, however, there are subtle differences in the performance of the pixels in the camera such that changes in either the lens or the illumination can result in changes in the gray values of the pixels output from the camera.

When the camera detects a light signal, the number of photoelectrons captured in each pixel during exposure is measured and transmitted to a computer as a digital grayscale value (DN). The conversion from electrons to DN follows a certain ratio, called the system gain (K), plus a fixed bias (usually 100 DN). These values are determined by a combination of the analog-to-digital converters and amplifiers used for the conversion. sCMOS cameras use parallel transmission, with one or more analog-to-digital converters in each column of the camera and an amplifier for each pixel, which results in small variations in gain and bias from pixel to pixel.

In dark field or low light conditions, the difference in bias can be measured by the DSNU mentioned in Section 3.7. Whereas in bright environments, the effect of gain also needs to be taken into account, and the difference due to gain and bias variations is measured by the photoresponse non-uniformity (PRNU), which is the ratio of detected electrons to displayed DN. Given that the difference in the resulting intensity values will depend on the size of the signal, it is expressed as a percentage.

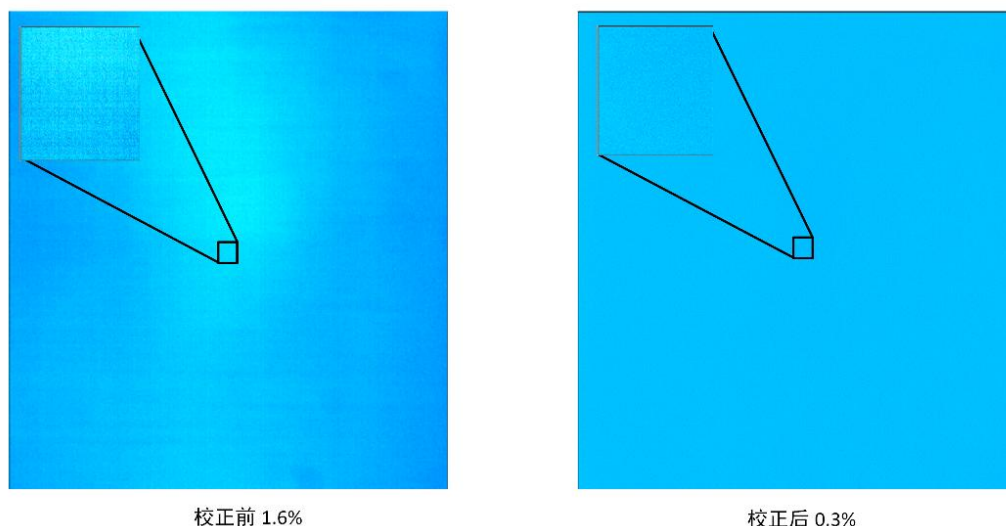


Figure3 -2 Comparison before (left) and after (right) PRNU correction

Typical PRNU values are <1%. For all low light and moderate intensity light source imaging (signals of 1000 e- or less), this variation is insignificant compared to read noise and other noise sources. Similarly, when imaging at high light levels, this variation is less significant compared to other noise sources in the image (e.g., optical scattering noise). However, in high light level imaging applications where very high measurement accuracy is required, especially those using frame averaging or frame summation, a PRNU value of <1% is highly desirable.

3.6. Region of Interest Readout

In imaging applications, ROI (Region of interest) is defined as a sub-region of interest within the resolution range of the camera sensor, and when the ROI is selected, only the images within this sub-region are read out. Note that the row windowing needs to be an integer multiple of 1 and the column windowing needs to be an integer multiple of 32.

Dhyana 2100 Camera ROI Area Measured Frame Rate (fps) Table

Resolution	Image Mode	Resolution			Frame Rate (Hz)		
FULLMode	Normal	OffsetX	Width	Height	8bit	10bit	12bit
		0	5120	4096	450	250	250
		0	5120	2048	885	600	494
		0	5120	1024	1712	1201	968
		0	5120	512	3217	2403	1856
		0	5120	256	5738	4806	3427
		0	5120	128	9434	9434	5943
		0	5120	64	13914	13914	9389
		0	5120	32	18248	18248	13224
		0	5120	16	18248	18248	13224
		0	5120	1	18248	18248	13224
		0	128	1	18248	18248	13224
	FastBinning	0	2560	2016	/		975
		0	2560	1013			1829
		0	2560	512			3338
		0	2560	256			5682
		0	2560	128			8755
		0	2560	64			11999
		0	2560	32			14727
		0	2560	16			14727
		0	2560	1			14727
		0	128	1			14727

Resolution	Image Mode	Resolution			Frame Rate (Hz)		
BaseMode	Normal	OffsetX	Width	Height	8bit	10bit	12bit
		0	5120	4096	225	150	150
		0	5120	2048	448	300	300
		0	5120	1024	888	600	600
		0	5120	512	1714	1201	1201

		0	5120	256	3094	2403	2403
		0	5120	128	5181	4806	4806
		0	5120	64	7816	7816	7575
		0	5120	32	10482	10482	10757
		0	5120	16	10482	10482	10757
		0	5120	1	10482	10482	10757
		0	128	1	10482	10482	10757
	FastBinning	0	2560	2016	/		610
		0	2560	1013			1201
		0	2560	512			2403
		0	2560	256			4548
		0	2560	128			7054
		0	2560	64			9735
		0	2560	32			12019
		0	2560	16			12019
		0	2560	1			12019
		0	128	1			12019

Note: The test frame rates in the table above are the maximum values measured at the shortest exposure time:

1) The test frame rate in the above table is the measured maximum value under the shortest exposure time .

2) The test environment is as follows:

Processor: Intel Xeon Platinum 8336C @ 2.30GHz

Motherboard: Supermicro X12SPA-TF

Memory: 192 GB (DDR4 3200MHz)

Primary Hard Drive: HP SSD FX900 Plus 2TB (2048 GB / SSD)

Graphics: ASPEED ASPEED Graphics Family (Supermicro)

Monitor: HAN2400 HD240 (24 inches)

Sound Card: Renesas ALC888 @ Intel High Definition Audio Controller

Network Card: Intel I210 Gigabit Network Connection / Supermicro

3.7. Incident Photon Calculation

Scientific camera imaging is a conversion process of photons, electrons, voltages, and gray values. Therefore the number of incident photons can be deduced inversely from the gray scale value. The calculation formula is shown below:

$$P = \frac{(DN - Offset)/K}{Q(\lambda)}$$

Where P represents the number of incident photons. DN is the gray value of the optical signal. K is the system gain in (DN/e-). $Q(\lambda)$ corresponds to the quantum efficiency when the light wavelength is λ . and Offset is the background value of the camera in DN.

3.8. Cooling

The camera is cooled to minimize the effects of "dark current noise" and hot pixels. The camera adopts semiconductor cooling, utilizing the Peltier effect, which consists of a pair of thermocouples made of N and P type materials. When the thermocouples are fed with DC current, heat absorption and exothermic phenomena are generated at the junction due to the different directions of the DC current flow. The cold end is close to the sensor to reduce the dark current. The hot end is connected to the metal heat-conducting block to disperse the generated heat.

Air-cooling and water-cooling are two common methods of heat dissipation. In air-cooling, the fan utilizes airflow to exchange excess heat with the surrounding air. In water-cooling, a liquid circulation system is used to transfer excess heat. The Dhyana 2100 camera supports both air-cooling and water-cooling, so users can choose the appropriate cooling method according to the actual usage environment.

The camera fan can be controlled at variable speeds. The higher the fan speed, the better the cooling performance. The lower the speed, the lower the transmitted vibration. For low vibration performance, the Dhyana 2100 camera's fan can be completely turned off in software and a liquid cooling port is provided (see Section 4.3 for water cooling installation).

4. Installation

4.1. Recommended Computer Configuration

Recommended servers that can run at full fps with dual card mode

Recommended configuration: Win10/ Core i7-13700 and above + 64GB of RAM

4.2. Camera Installation

1. **Power supply connection:** Use the 24V/6.67A power supply from the factory.
2. **Water-cooling system connection** (if the camera is air-cooled working mode, please omit this step): use the recommended supporting water circulation system to connect the camera's two water ports, open the water circulation, and ensure that the water circuit is smooth and does not leak. The recommended water temperature 25 °C, 20 °C.
3. **Frame grabber driver installation:** supports the Euresys frame grabber, computer installation of the frame grabber (recommended installation to PCIE 3.0 x16 slot) and install the frame grabber supporting the driver:

Use the Euresys driver version for egrabber-win10-x86_64-23.10.3.77 for frame grabber driver installation. For two frame grabbers, the firmware version is upgraded to "1-camera". When the upgrade is complete, on the desktop, click "Shutdown" and power on the computer again. The upgrade is now successful.
4. **CXP connection:** use the factory matching CXP-12 cable to connect the camera end and frame grabber end, single frame grabber connected to 4 cables (need to connect to the host, can not be connected to the slave), dual frame grabber connected to 8 cables, each frame grabber is connected to lock the wiring port of each cable.
5. **Camera power-up:** wait until the camera startup is complete, open the supporting software. if water-cooled alone, you must first turn on the water circulation system and

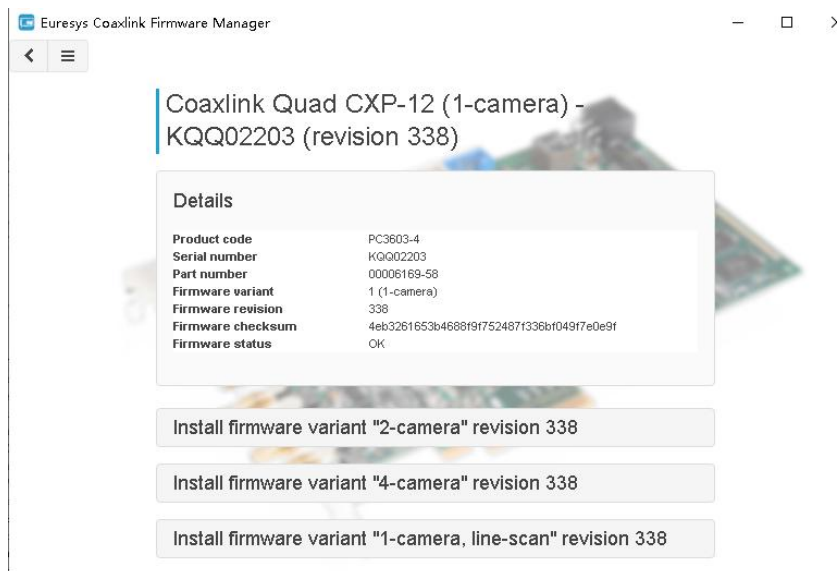
normal operation, and then turn on the camera, so as not to overheat the camera.

6. Precautions

(1) When connecting 4 CXP cables to each frame grabber, the wires should be in consecutive and one-to-one correspondence.

(2) Steps to switch the firmware mode of Euresys frame grabber: Open Coaxlink Firmware Manager in the driver installation directory -> select the firmware -> burn successfully -> turn off the computer -> turn on the computer -> the frame grabber firmware takes effect.

Note: The firmware takes effect when the computer is turned off and powered on again instead of restarting the computer.



4.3. Water-cooled tube installation

4.3.1. Connecting the water cooling tubes

Procedure.

- (1) Place the imager on a stable workbench.
- (2) Connect the water-cooling hose to the water hose connector on top of the imager,

making sure it is inserted in place.

(3) Insert the water hose into the nozzle of the cooling water circulator and tighten it with the clamp .

(4) The cooling water circulator hose is connected to the imager hose via an adapter valve.

4.3.2. Software Fan Status

(1) After installing the water-cooling tubes, you need to switch the cooling method on the software. The camera's default cooling method is air cooling, which can be switched to off by adjusting the fan stop.

(2) If you choose to turn off the fan, please make sure the water cooling system can work normally before doing this operation.

4.3.3. Disconnecting the water cooling tubes

(1) Disconnect the camera and all other equipment including the water cooler circulator.

(2) Drain the water from the circulator by unplugging the water hose above the cooling water circulator according to the instructions for the circulator.

(3) Press the adapter valve slide to unplug the cooling water circulator hose and drain the water inside.

(4) Press on the water fitting to pull the camera water hose out of the water fitting. First, with the side where the water valve is mounted facing sideways (**not overhead**), and with the water valve opening facing down when pulling out the water hose, use an absorbent towel or paper towel for good protection to ensure that no water leaks into the camera.

Attention:

1) Selection of cooling water: Deionized water is recommended.

2) Water Temperature: Generally, the recommended water temperature is 20 °C . Inappropriate environmental conditions can cause condensation on the water valve and water pipes, which may damage the equipment. In order to ensure the normal operation of the equipment, it is recommended that the water temperature should not be lower than the

condensation point, the condensation point please refer to the appendix condensation table.

3) *Pressure: The maximum water pressure entering the camera should not exceed 2 bar.*

4) *Cooling water circulator: Use the circulator as well as the chilled water properly according to the instructions for the cooling water circulator.*

5) *Install the water cooling lines correctly and make sure that there is no water seepage at the interface between the camera end and the cooling water circulator end.*

6) *Before running the camera, make sure that the cooling water circulator and the water valve of the camera are installed correctly, and make sure that the water flow of the water pipe reaches 1L/min.*

7) *At the end of the operation, turn off the power of the camera as well as the circulator and other related equipment, and drain the water from the circulator and the camera.*

Warning:

1) ***During the operation of the camera, do not stop cooling by turning off the cooling water circulator. Failure to do so may result in damage to the sensor from continued high temperature operation.***

2) ***During imager operation, water cooling and fan can be turned on at the same time, but cannot be turned off at the same time, at least one type of cooling is required to operate normally, otherwise it may lead to damage of the sensor by sustained high temperature operation.***

4.4. frame grabber installation

Turn the computer off and open the cover of the computer mainframe as shown in Figure 4-6 . Select a PCIe slot with a transmission bandwidth greater than 850MB/s to insert the frame grabber and secure it with screws before restarting the computer. Connect the camera to the PC frame grabber via the data transfer cable.

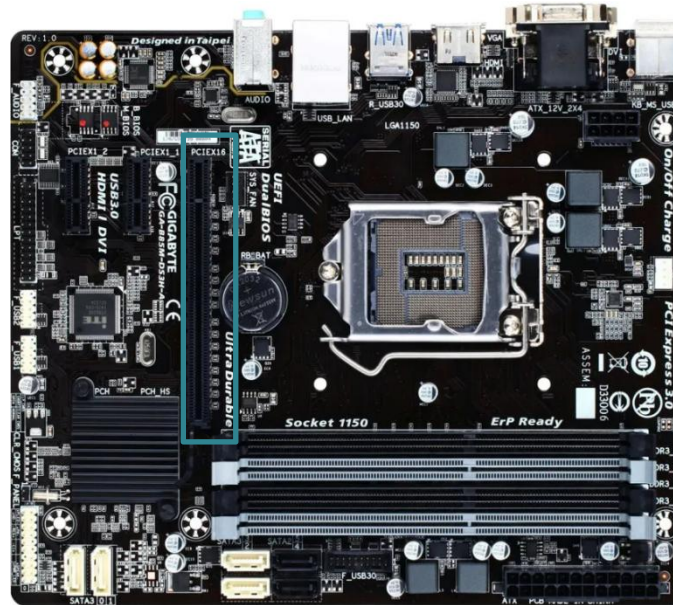


Figure -4 1 Computer motherboard diagram

Table -41 Maximum transfer rates for different PCIe slots

PCIe	X1	X4	X8	X16
1.0	250MB/s	1GB/s	2GB/s	4GB/s
2.0	500MB/s	2GB/s	4GB/s	8GB/s
3.0	985MB/s	3.9GB/s	7.8GB/s	15.7GB/s

Caution:

- 1) When installing as well as removing the frame grabber, be sure to operate with the power off.
- 2) When connecting the CXP cable, please operate with the camera's power off, and prohibit hot plugging and unplugging.

4.5. Driver Installation and Uninstallation

This section describes the driver installation and uninstallation procedures for the frame grabber.

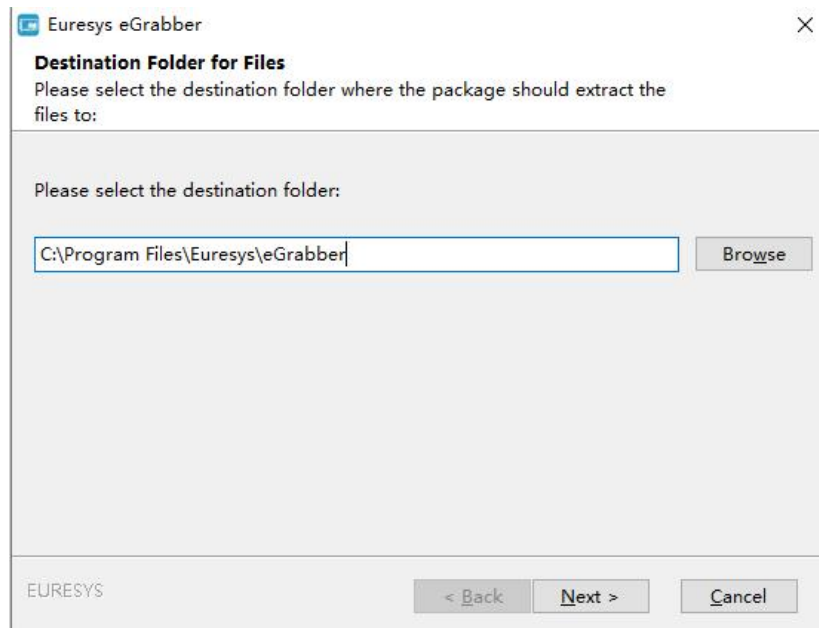
4.5.1. Euresys frame grabber Driver Installation

Note: Euresys frame grabber in Windows supports Win10 system.

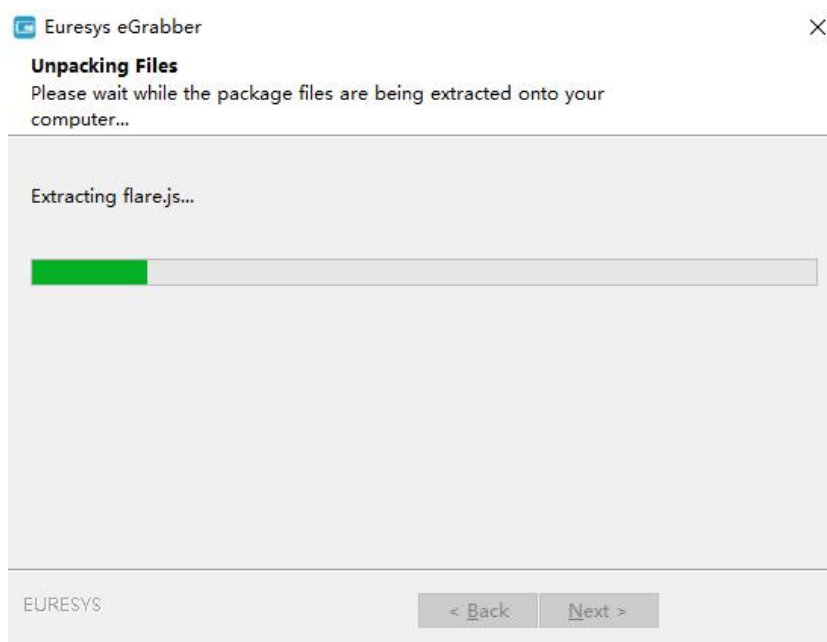
Technical Support Email: service@tucsen.com TEL: +86-591-28055080-818 Fax: +86-591-28055080-826

egrabber-win10-x86_64-23.10.3.77

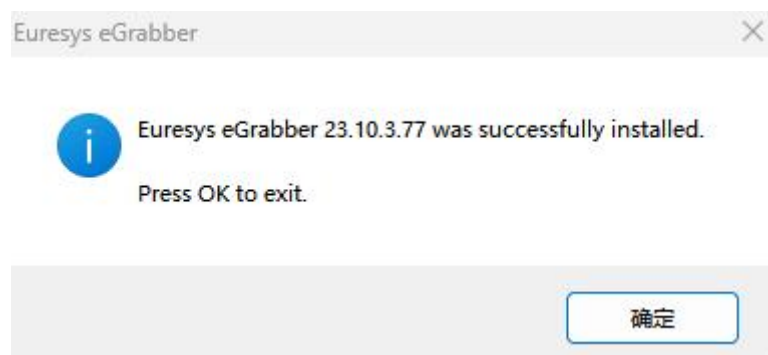
(1) Take Win10 as an example, double-click the corresponding driver to start the installation, and click "Next" for the next step according to the default.



(2) Wait for the installation to complete



- (3) Click "Confirm" to confirm the installation is complete.



4.6. Software Installation

The Samplepro software comes with the USB flash drive, unzip it directly to use it.

Please note the following:

- (1) The first time you start the Samplepro software, run it as right-click administrator, and then double-click the software directly.
- (2) If the frame grabber software and Samplepro software are opened at the same time, the software will not recognize the camera.

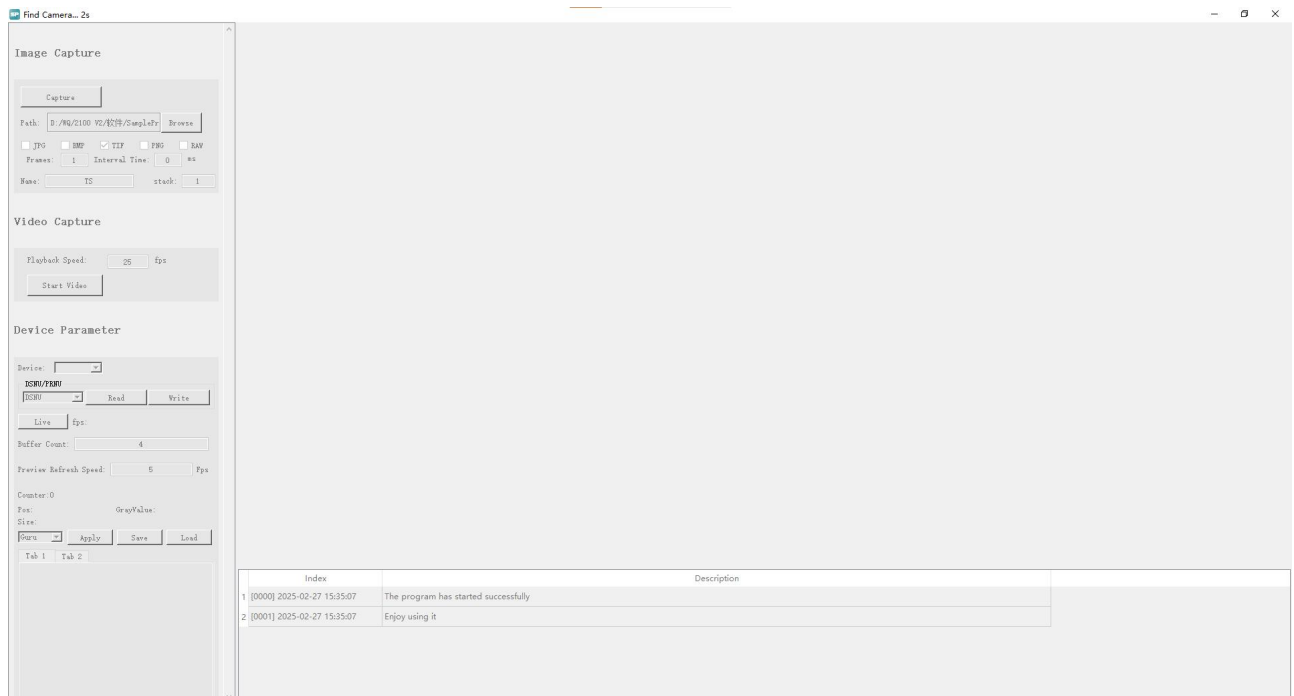
The green light is always on when you open the camera, and the green and orange lights flash alternately when the camera is out of the stream. If the startup software NO Camera, please check the camera and frame grabber wiring order is matched, or restart the computer, reinstall the frame grabber driver to try.

5. Instructions for using the software

5.1. Software startup interface

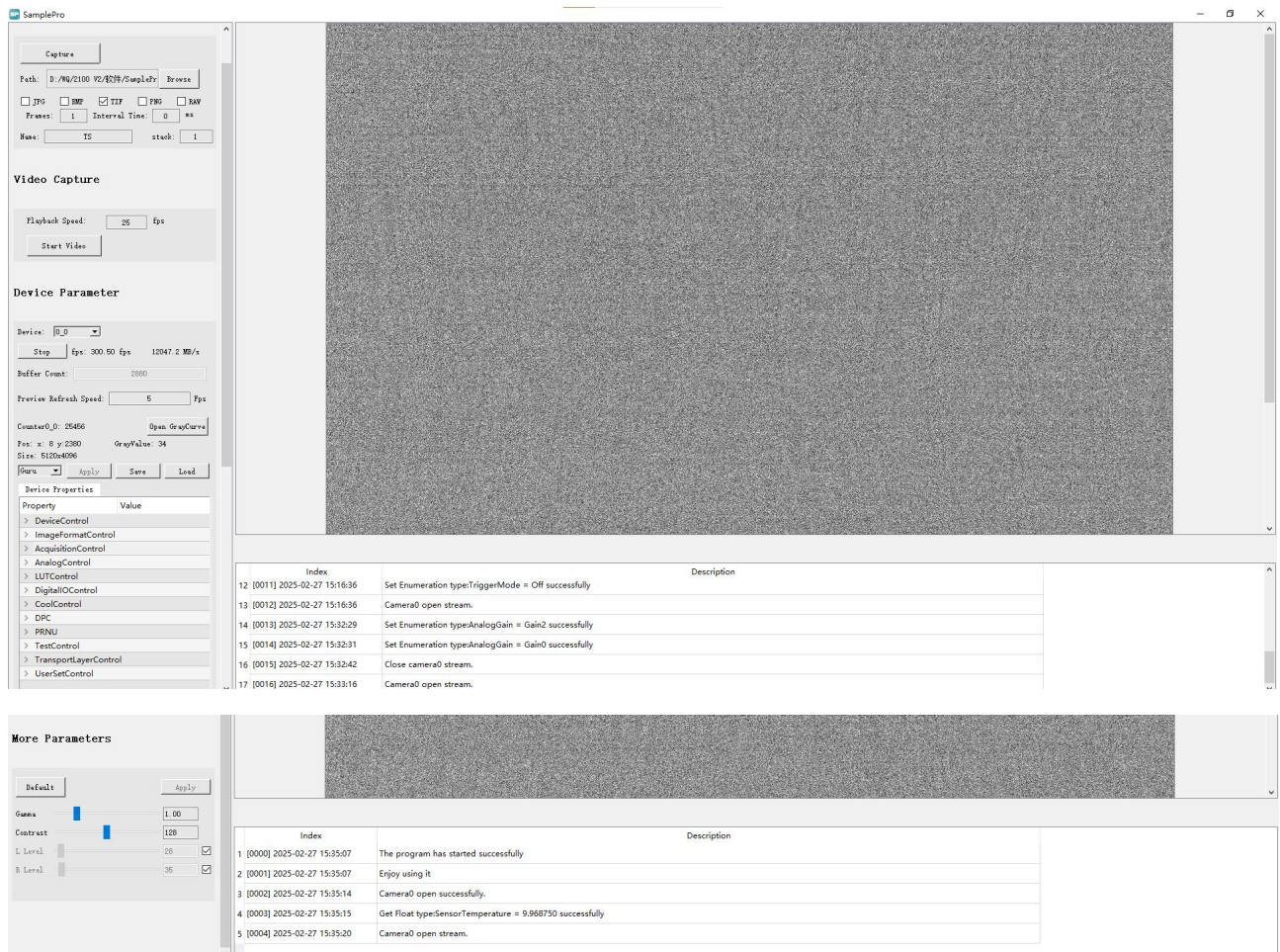
Wait for about 20s after the camera is powered on, double-click SamplePro to open the camera software, the software startup interface as shown in the following figure , wait for the loading to be completed, using different frame grabbers with different startup time (Euresys frame grabber startup time is about 5s, Samadhi frame grabber startup time is

about 10s).



5.2. Window Composition

The main interface of SamplePro software consists of "Preview Window", "Software Window", "Image Camera", "Camera Control", and "Image Control". "and" image adjustment "five parts, as shown below.



5.2.1. Preview Window

The preview window displays the real-time shooting screen of the camera under the stream module. The preview window supports real-time zoom, so users can zoom in or out of the preview window with the mouse wheel according to their actual needs.

The left side of the preview window displays the grayscale value, coordinates and resolution of the image where the mouse is located in the live preview screen, as shown in the figure below.

```
Counter0: 12818
Pos: x: 288 y: 1240      GrayValue: 124
Size: 5120x4096
```

5.2.2. Software Window

The software window is the common " - " window minimization function, " □ " - maximize, " ✕ " - close window function.

5.2.3. Image Camera

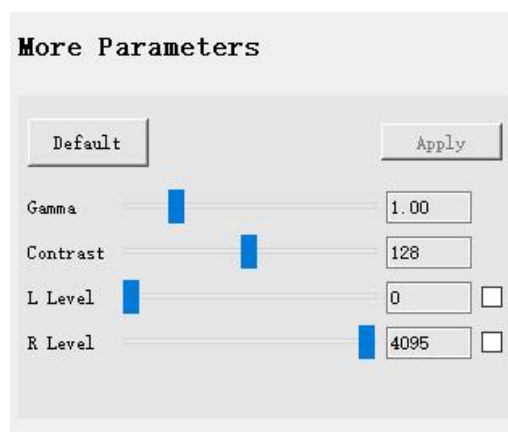
The image photography module is the basic picture taking and video recording function of the camera. Users can choose different image formats to capture and customize the duration of video recording according to the usage requirements, which will be saved in the img folder in the root directory of the software by default.

5.2.4. Camera Control

The main functions of the camera are carried out under this module, and the user can carry out the use under the corresponding function module according to the use requirements.

5.2.5. Image Adjustment

In the Image Adjustment interface, users can adjust the image gamma, contrast, and color gradation according to the difference between the real-time preview effect and the actual sample, in order to achieve the desired image effect.



5.3. Image Camera

As shown in the figure below, the image camera module supports the basic functions of the camera to take pictures and videos.

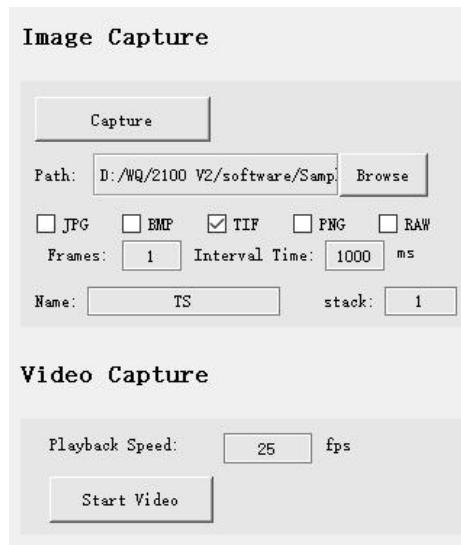


Image Capture

Capture

Path: D:/WQ/2100 V2/software/Samp Browse

☐ JPG ☐ BMP ☒ TIF ☐ PNG ☐ RAW

Frames: 1 Interval Time: 1000 ms

Name: TS stack: 1

Video Capture

Playback Speed: 25 fps

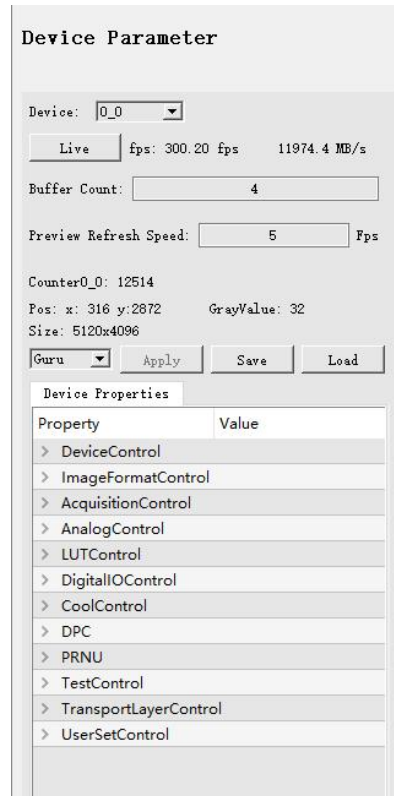
Start Video

- ◆ Capture: Click the button to take a picture.
- ◆ Path: Capture the image and video save path, click Browse to customize the save path.
- ◆ ☐ JPG ☐ BMP ☒ TIF ☐ PNG ☐ RAW Image format: support JPG, BMP, TIF, PNG and RAW, the default is TIF format, support for multiple image formats at the same time check the box.
- ◆ Frames: customize the number of pictures, default is 1 picture.
- ◆ Interval Time: interval time, default 1000ms.
- ◆ Name: name of the picture, support customization, default TS prefix.
- ◆ stack: select TIF to store pictures, default is 1, if the setting is not 1, then MPTIF will be generated.
- ◆ Playback Speed: set the frame rate of video playback, default 25fps.
- ◆ Start Video: manually start video recording.
- ◆ Stop Video: manually stop video recording.

5.4. Camera Control

The main functions of the camera are found in this module. The camera's image mode

is controlled here, as shown in the following figure.



- ◆ Stop: the camera is in streaming mode, the preview window will show images in real time and, in external trigger mode, the images will be shown according to the trigger signal.
- ◆ Live: the camera is working normally, no new images are displayed.
- ◆ Buffer Counter: apply the number of stored memory cache frames. It is recommended to set 70% of the total memory size when you need to store a large number of images.
- ◆ Preview Refresh Speed: set the preview frame rate of real-time data stream, when not focusing on the image content can be set to 0 to close the preview display, the maximum preview frame rate does not exceed the camera's real-time frame rate.
- ◆ fps: display the real-time frame rate of the camera.
- ◆ Counter: the cumulative number of preview frames.



- ◆ Operator Mode: operator mode selection, divided into Beginner, Expert,

Guru. To use all the functions of the camera, please select the Guru mode.

5.5. DeviceControl

Display camera basic information (read-only), UserID editing, camera reset, Device temperature, Sensor Board temperature, Sensor temperature real-time display, timestamp display, reset. As shown in the figure below.

Device Properties	
Property	Value
▼ DeviceControl	
DeviceMode	FULL Mode
DeviceSlaveState	Successful
DeviceVendorName	Tucsen
DeviceModelName	Dhyana 2100
DeviceManufacturer	Tucsen CXP Camera
DeviceVersion	1701250217
DeviceSerialNumber	123123
DeviceUserID	
DeviceSFNCVersion	2
DeviceSFNCVersion	0
DeviceSFNCVersion	0
DeviceManifest...	0
DeviceManifest...	1
DeviceManifest...	0
DeviceManifest...	4
DeviceManifest...	1
DeviceManifest...	0
DeviceManifest...	Local:Dhyana_210...
DeviceTLType	Coa X Press

Property	Value
DeviceManifest...	Local:Dhyana_210...
DeviceTLType	Coa X Press
DeviceTLVersion...	1
DeviceTLVersion...	1
DeviceTLVersion...	0
DeviceLinkSelect...	0
DeviceLinkSpeed	0.00000
DeviceLinkThro...	0.00000
DeviceLinkCom...	0.00000
DeviceReset	{Command}
DeviceIndicator...	Active
DeviceTempera...	58.51590
SensorTemper...	9.96875
Timestamp	68,205,989,961,184
TimestampIncr...	8
TimestampReset	{Command}
TimestampRes...	0
TimestampLatch	{Command}
TimestampLatc...	0
SensorFrameC...	31

- ◆ DeviceMode: Switch between single and dual frame grabber channels of the camera. Select Base Mode when the camera uses a single frame grabber. Select FULL Mode when it uses dual frame grabbers.
- ◆ DeviceUserID: DeviceUserID supports customized input editing, supports Chinese and

English and other commonly used numeric character input.

- ◆ DeviceReset: device reset button. There is no need to power off the camera as the camera automatically loads the default parameter configuration.
- ◆ DeviceTemperature: real-time display of Device temperature (read-only).
- ◆ SensorTemperature: real-time display of Sensor temperature (read-only).
- ◆ Timestamp: Timestamp information display, precision 8 ns (read-only).
- ◆ TimestampReset: timestamp reset. Recount timestamp from the current intercepted frame.
- ◆ TimestampLatch: capture current timestamp information.

Timestamp	1,952,597,131,304
TimestampIncr...	8
TimestampReset	{Command}
TimestampRes...	0
TimestampLatch	{Command}
TimestampLatc...	0
SensorFrameC...	459,880
CxpFrameCoun...	229,938

- ◆ SensorFrameCounter: current Sensor count, total data sent.
- ◆ CxpFrameCounter: current CXP end count. When the camera is dual camera mode, the CXP end count is half of the Sensor.

Note: when the frame rate is too high, there will be a few frames data sent out from Sensor, but CXP has stopped counting, there will be a case of losing a few frames.

5.6. ImageFormatControl

Set camera resolution, ROI, Binning Mode, Horizontal and Vertical Mirror, Pixel Width, Test Image. As shown in the figure below.

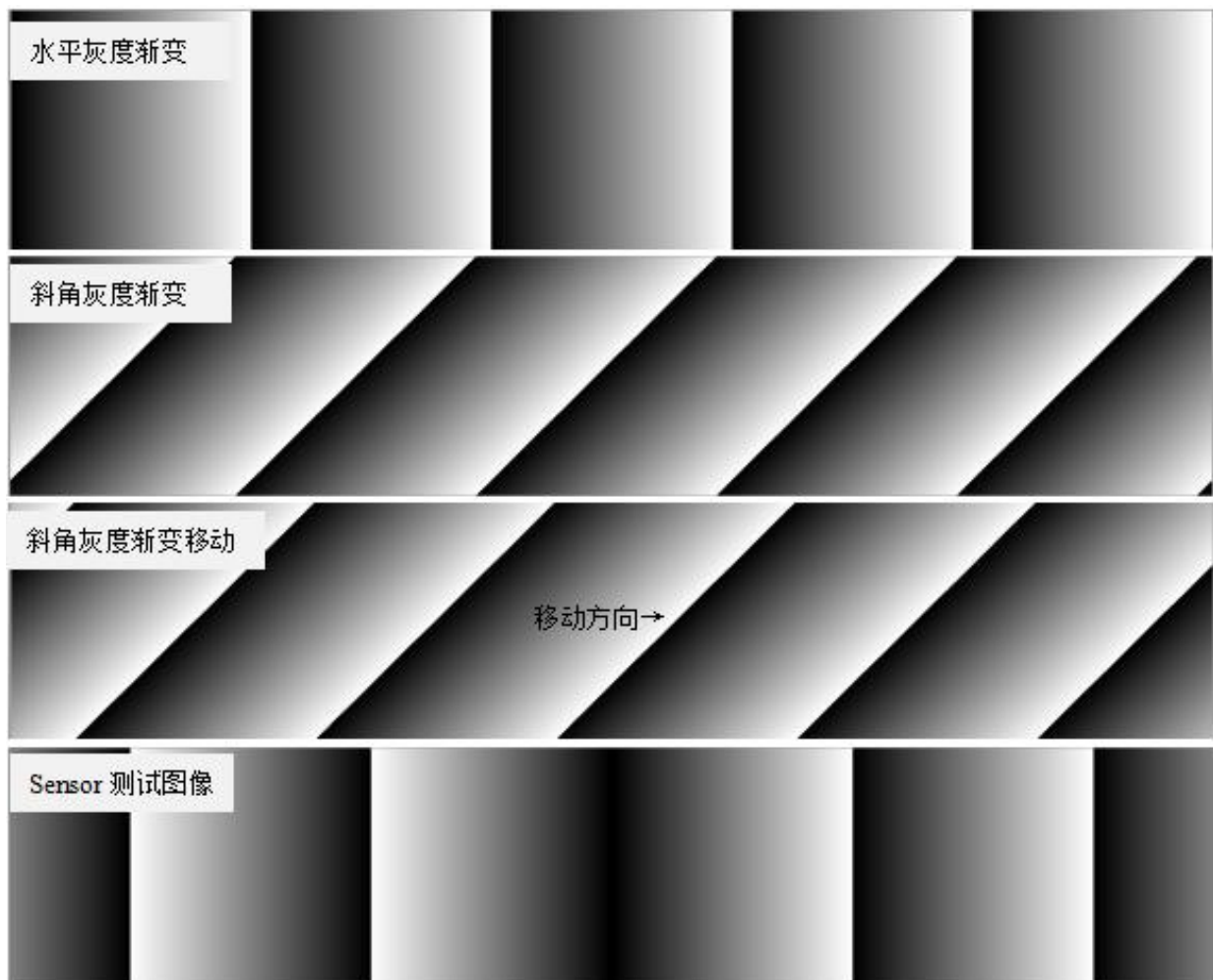
ImageFormatControl	
SensorMode	Normal
WidthMax	5,120
HeightMax	4,096
Width	5,120
Height	4,096
OffsetX	0
OffsetY	0
BinningAver	☒ True
BinningHorizon...	X1
BinningVertical	X1
ReverseX	☐ False
ReverseY	☐ False
PixelFormat	Mono 12
TimestampEn	☒ True
TestPattern	Off

- ◆ **SensorMode:** The sensor supports Normal and FastBinning modes, the camera frame rate and signal-to-noise ratio can be improved under FastBinning (only 12bit supports FastBinning mode).
- ◆ **WidthMax:** real-time display of the maximum horizontal resolution of the current image (gray read-only).
- ◆ **HeightMax:** real-time display of the maximum vertical resolution of the current image (gray read-only).

Note: Normal resolution of 5120 × 4096, FastBinning resolution of 2560 × 2016

- ◆ **Width:** current image horizontal width setting, minimum input 128, step 32.
- ◆ **Height:** current image height setting, minimum input 1, step 1.
- ◆ **OffsetX:** the current image horizontal width starting point position, minimum input 0, step 1.
- ◆ **OffsetY:** current image height start point position, gray read-only, FPGA automatically centered according to Height setting after ROI to improve frame rate. Example: Under Normal, Height=2048, then OffsetY=1024.
- ◆ **BinningAver:** camera Binning mode switching. When enabled, Binning is Average Bin. When not enabled, Binning is SumBin.
- ◆ **BinningHorizontal:** Horizontal Binning mode selection, support x1, x2, x4, x8.

- ◆ BinningVertical: Vertical Binning mode selection, support $\times 1$, $\times 2$, $\times 4$, $\times 8$.
- ◆ ReverseX: Horizontal Mirror switch, check the box to flip the image horizontally.
- ◆ ReverseY: Vertical Mirror switch, check the box to flip the image vertically.
- ◆ PixelFormat: switch pixel width. Supports 8bit, 10bit, and 12bit.
- ◆ TimestampEn: timestamp statistics switch. When enabled, the gray value of 1~14 pixels in the first row of the upper left corner of the image will be used to count the time of the current frame, the number of frames, the start of the readout and the start of the exposure information.
- ◆ TestPattern: Sensor raw data output. Supports horizontal grayscale gradient, diagonal grayscale gradient, diagonal grayscale gradient shift, Sensor test image. as shown below.



5.7. AcquisitionControl.

Set camera frame rate, trigger mode, exposure mode, etc.. as shown in the following figure.

AcquisitionControl	
AcquisitionMode	Continuous
AcquisitionStart	{Command}
AcquisitionStop	{Command}
AcquisitionFra...	250.26387
AcquisitionFra...	250.26387
FrameRateMode	On
ExposureMode	Timed
ExposureTime...	3.98078
ExposureTime	3.98078
TriggerMode	Off
TriggerFrameN...	1
TriggerSoftware	{Command}
TriggerSource	ExternalTrigger
TriggerActivation	Rising Edge

- ◆ AcquisitionStart: in streaming mode, start to continue counting after pausing the image.
- ◆ AcquisitionStop: in streaming mode, pause the current software and stop counting.
- ◆ AcquisitionFrameRateMax: display the maximum frame rate that can be set by the camera under the current parameters (gray read-only).
- ◆ AcquisitionFrameRate: set the current frame rate of the camera. The minimum setting is 0.095Hz.
- ◆ FrameRateMode: in the On state, the frame rate setting is prioritized, at this time the maximum settable exposure time $s=0.95/\text{frame rate}$. In the Off state, the exposure time setting is prioritized, at this time the maximum settable frame rate $\text{Hz}=0.95/\text{exposure time } s$.
- ◆ ExposureMode: Exposure mode selection. Supports Timed (manual exposure time is determined by the interface parameter settings), Auto (automatic exposure mode, the maximum automatic exposure time of 50ms), PulseWidth (exposure time is determined by the duration of the high level of the external trigger) three modes

Note: 1. PulseWidth is selectable when TriggerMode is On.

2. The external trigger mode does not support auto exposure, when the external trigger mode is switched to auto exposure, it will trigger the internal trigger out of the picture.

- ◆ ExposureTimeMax: displays the maximum exposure time that can be set by the camera under the current parameters (gray read-only).
- ◆ ExposureTime: set the current exposure time of the camera, unit ms. Manual exposure time range 4us~10s, automatic exposure time range 4us~50ms.
- ◆ TriggerMode: Trigger mode, On state opens external trigger, exposure mode can choose PulseWidth.

Note: When using Full Mode with dual frame grabber, the number of external triggers must be a multiple of 2.

- ◆ TriggerFrameNum: Set how many frames to be triggered by one pulse of external trigger and soft trigger. For example, maximum frame rate of external trigger camera is 100Hz, pulse frequency is set to 50Hz, the number of pulses is 100, TriggerFrameNum is set to 2, and the frame rate of the camera will be doubled when the trigger signal is triggered to 100Hz, triggering 200 frames out of the image.

Note: External trigger mode, TriggerFrameNum recommended to use the default parameter 1, TriggerFrameNum>1 may cause the trigger frame loss.

- ◆ TriggerSoftware: software trigger, click the button to trigger the image.
- ◆ TriggerSource: support ExternalTrigger and SoftTrigger, when TriggerMode is On, users can choose ExternalTrigger and SoftTrigger according to the triggering demand.
- ◆ TriggerActivation: Trigger polarity selection, supports Rising Edge (valid on rising edge), Falling Edge (valid on falling edge), Any Edge (valid on both upper and lower edges).

Note: When ExposureMode is in PulseWidth, the trigger polarity does not support Any Edge, when selecting the dual frame grabber mode, the number of triggers must be a

multiple of 2, if it is odd, it will lead to frame loss.

5.8. AnalogControl

Analog, digital gain, black level adjustment as shown in the figure below.

▼ AnalogControl	
AnalogGain	Gain0
DigitalGain	X1.0000
BlackLevel	0

- ◆ AnalogGain: Analog gain. Supports Gain0, Gain1, Gain2, Gain3.
- ◆ DigitalGain: digital gain. Supports $\times 1 \sim \times 15.5$ switching.
- ◆ BlackLevel: black level. Increase or decrease the image background gray value, 8bit adjustable -255~255, 10bit and 12bit under the proportional conversion.

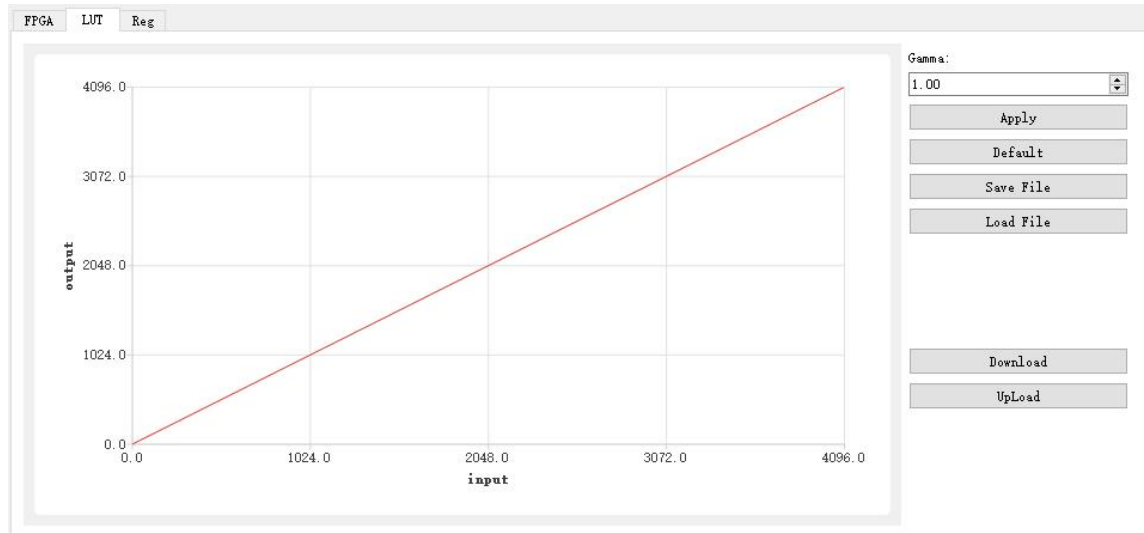
5.9. LUTControl

LUT module is shown in the following figure.

▼ LUTControl	
LUTSelector	Luminance
LUTEnable	☐ False
LUTIndex	0
LUTValue	0
LUTSave	{Command}
LUTLoad	{Command}

- ◆ LUTEnable: LUT control switch, not on by default.
- ◆ LUTIndex: INPUT value, the output value corresponding to the applied curve is automatically loaded after input, range 0~4095.
- ◆ LUTValue: OUTPUT value, loaded according to the input LUTIndex value, can be manually modified and saved, range 0~4095.
- ◆ LUTSave: save the modified curve.
- ◆ LUTLoad: load the LUT curve written by the application.

Before using LUT function, users need to open UpdateFW software to write LUT configuration according to the actual use, as shown in the following figure.



- ◆ Gamma: Input the value to adjust the corresponding INPUT-OUTPUT curve.
- ◆ Apply: Apply the currently called LUT curve.
- ◆ Default: Restore the default LUT curve state when Gamma=1.
- ◆ Save File: save the current LUT curve to the specified file path.
- ◆ Load File: load the saved LUT curve file.
- ◆ DownLoad: configure the currently applied LUT curve to the camera.
- ◆ UpLoad: load and read the applied LUT curve from the camera (you need to save the LUT curve in SamplePro before you can load and read it).

The application steps are as follows: 1. Enter the required gamma value. 2. Click Apply to apply the currently selected curve. 3. Click DownLoad to configure the curve into the camera.

5.10. DigitalIOControl

The camera trigger output module is shown below.

▼ DigitalIOControl	
TriggerOutSel	Ch1
▼ TriggerOutConfig	
TriggerOut...	ExposureStart
TriggerOutl...	☐ False
TriggerOut...	0
TriggerOut...	10

- ◆ TriggerOutSel: two output channels Ch0 and Ch1 are selected.
- ◆ TriggerOutMode: select the output type for the currently selected Ch0 or Ch1, both output channels support ExposureStart (exposure start signal), ReadEnd (read out end signal).
- ◆ TriggerOutInverter: high bottom level inversion enable, control output signal inversion.
- ◆ TriggerOutDelay: output signal delay exposure time, adjustable range 0~10000, unit us.
- ◆ TriggerOutDuration: output signal high level duration, adjustable range 0~10000, unit us.

5.11. CoolControl

The camera cooling control module is shown in the figure below.

▼ CoolControl	
FanOperationM...	Temperature
FanDutyFactor	40
TECOperation...	Temperature
TECDutyFactor	75
CoolOperation...	10

- ◆ FanOperationMode: fan mode gear, Off (off fan), On (on fan), Temperature (constant temperature mode).
- ◆ FanDutyFactor: fan speed, adjustable from 20~100%.
- ◆ TECOperationMode: TEC mode, Off (fan off), On (fan on), Temperature (constant temperature mode).

- ◆ TECDutyFactor: TEC power factor, adjustable 0~100%.
- ◆ CoolOperationTemperature: cooling target temperature, adjustable range -30~50℃, default 10℃.

5.12. DPC

▼ DPC	
DPCMode	On
DPCLevel	Low
DPCGenerate	{Command}
DPCSave	{Command}
DPCLoad	{Command}

- ◆ DPCMode: Static Defective Pixel Correction enable switch Off, On.
- ◆ DPCLevel: static Defective Pixel Correction gear selection, only supports Low.
- ◆ DPCGenerate: click DPCGenerate for correction under long exposure time in dark field environment.
- ◆ DPCSave: save the DPCGenerate calibration results.
- ◆ DPCLoad: load the Defective Pixel Correction result.

5.13. PRNU

▼ PRNU	
PRNUMode	On
PRNUSelector	PRNU 0
TargetLevelAUTO	☒ True
PRNUTargetLevel	200
PRNUGenerate	{Command}
PRNUSave	{Command}
PRNULoad	{Command}
PRNUFactorySa...	{Command}
PRNUFactoryRe...	{Command}

- ◆ PRNUMode: PRNU calibration enable switch. Off position turns off PRNU, ON position turns on PRNU.
- ◆ PRNUSelector: PRNU gear selector. Supports PRNU0~PRNU5 total 6 groups of data.

- ◆ **TargetLevelAUTO**: automatic PRNU mode (the system automatically calculates the average value of the current image brightness), after checking the enable box, click PRNUGenerate under the uniform light of the bright field to automatically generate PRNU correction data.
- ◆ **PRNUTargetLevel**: manual PRNU mode. Uncheck the automatic PRNU mode, set the PRNU gray value, adjust the brightness of the bright field uniform light to the set gray value, then click PRNUGenerate to automatically generate PRNU correction data.
(Example: if the PRNUTargetLevel is set to 128, then adjust the bright field uniform light source to a roughly semi-saturated state, and then one-click online PRNU correction).
- ◆ **PRNUGenerate**: click to do online PRNU correction.
- ◆ **PRNUSave**: save the current PRNU result to the selected PRNU0~PRNU7.
- ◆ **PRNULoad**: load the saved PRNU data from the selected PRNU0~PRNU7.
- ◆ **PRNUFactoryReset**: restores the selected PRNU0~PRNU7 to the factory backup parameters.

Note: To switch between different bit depths, you need to use the corresponding PRNUSelector, because the PRNU backgrounds of different bit depths are different, and using mismatched PRNU backgrounds will lead to abnormal images. Example: If the current PRNUSelector is PRNU0 and the image parameter is set to FastBinning-12bit, the image preview will be abnormal because the Normal and FastBinning resolutions are different, resulting in incompatible backgrounds.

PRNUSelector correspondence:

PRNU0	Normal-10bit-Gain0
PRNU1	Normal-12bit-Gain0
	FastBinning-12bit-Gain0
PRNU2	Normal-10bit-Gain1
PRNU3	Normal-12bit-Gain1

	FastBinning-12bit-Gain1
PRNU4	Normal-10bit-Gain2
PRNU5	Normal-12bit-Gain2
	FastBinning-12bit-Gain2
PRNU6	Normal-10bit-Gain3
PRNU7	Normal-12bit-Gain3
	FastBinning-12bit-Gain3

5.14. CXP Switching

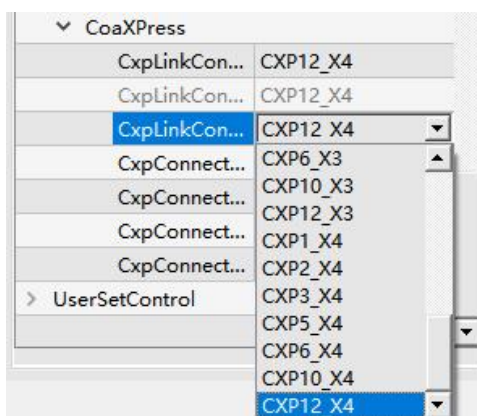


Figure 4-16.

- ◆ CxpLinkConfigurationPreferredSwitch: CXP default startup connection selection.
- ◆ CxpLinkConfiguration: CXP mode switching, default CXP12_X4, need to be switched in the live state, the stream mode Stop state gray can not be switched.

5.15. UserSetControl

The user configuration module is shown in the following figure.

UserSetControl	
UserSetSelector	User Set 1
UserSetLoad	{Command}
UserSetSave	{Command}
UserSetDefault	User Set 1

- ◆ UserSetSelector: user settings storage. Provides Default, User Set1, User Set2 three kinds of configuration.
- ◆ UserSetLoad: the camera loads the selected Default, User Set1, User Set2 set of parameters.
- ◆ UserSetSave: the configuration of the modified parameters is saved in the selected group of User Set1, User Set2.
- ◆ UserSetDefault: a group of user configurations loaded by default after the camera is reset or restarted by power failure, as shown in Figure 4-17, the User Configuration Module loads the settings of User Set1 by default.

6. Maintenance

Damage caused by unauthorized maintenance or procedures will void the warranty.

6.1. Periodic Inspection

The condition of the product should be checked periodically, especially the integrity of the external power supply and mains cables, the integrity of the power cord, and do not use damaged equipment.

6.2. Electrical safety check

- It is recommended that the integrity of the AC/DC converter's insulation and protective earth is checked annually
- Do not use damaged equipment

6.3. Coolant hoses and connections

Users should regularly inspect all coolant hoses and connections for signs of leakage, damage or wear. All seals must be intact before the imager system can be turned on and any worn or damaged parts must be replaced immediately.

6.4. Basic Use

- 1) Avoid opening the lens dust cap in a dusty environment.
- 2) Open the lens dust cap and mount the lens with the camera port facing downward to avoid dust falling on the surface of the window piece.
- 3) Attach the dust cap when not in use for a long time.

6.5. Window Blade Cleaning

When you find that the images captured by the camera have stains or dirty spots, please rule out whether the camera interface lens/microscope/objective lens, etc. is dirty on the surface.

If none of the above is true and it is determined that it is from the camera itself, you can follow the steps below for window cleaning:

- 1) Prioritize the use of air blowing, use earwash ball or air blowing to blow off general dust. together with the brush can remove most of the dust.
- 2) For stubborn oily dust, you need to prepare dust-free cotton swabs (or special mirror paper, non-woven fabric, etc.) and special tools such as anhydrous ethanol.
- 3) Use dust-free cotton swabs dipped in an appropriate amount of anhydrous ethanol to wipe along the window surface, wipe, do not wipe too hard, and always along a direction, avoid back and forth wipe.
- 4) After wiping well, use air blowers and other items to let the alcohol on the surface of the window evaporate completely before continuing to use.

Caution:

- 1) *If you cannot guarantee to complete the cleaning steps independently or do not have the required items, please be sure to contact us at .*
- 2) *If the dirt still exists after following the above steps, try wiping again by following the above steps. If this does not solve the problem, consider that the sensor is internally dirty. In this case, please contact us.*

7. Troubleshooting

7.1. The camera is not recognized by the computer

- 1) Make sure that the camera is powered on.
- 2) Confirm that the camera is normally connected to the computer and the driver is installed.

7.2. Software suspended work, stuck

- 1) If the computer has turned on energy-saving mode, the system CPU performance will be reduced, resulting in the software not working properly, such as dropped frames or software jamming. Check to ensure that the computer is in high performance mode.
- 2) The computer has opened too many applications, resulting in the computer CPU occupancy being too high. If the software CPU utilization is low, the software cannot work properly. Close any redundant applications.
- 3) Data cable connection abnormality. When the data cable is loose, or extended by the adapter too long, this can lead to software connection abnormalities.

7.3. The camera does not reach the target cooling temperature

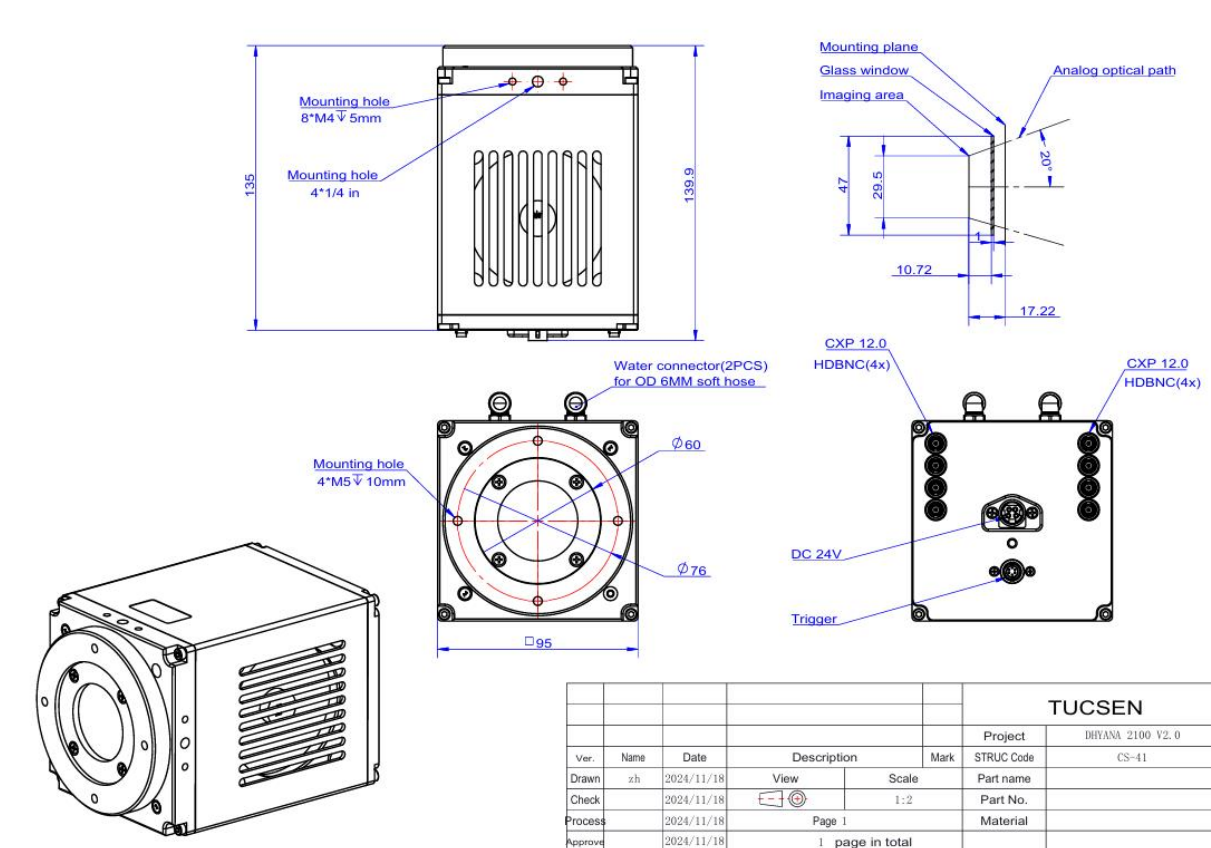
- 1) Confirm the ambient temperature and that the camera has the maximum cooling temperature difference.
- 2) Confirm whether the air outlet is blocked.
- 3) Confirm whether the fan rotates normally.
- 4) If you use water-cooling, please confirm whether the water-cooling cycle is running normally.

8. After-sales Support

- 1) Login to the official website and click [Technical Support] module to get FAQ.
- 2) Warranty:
 - The product warranty period is 24 months from the date of shipment. Any damage during this period will be repaired free of charge if it meets the warranty requirements.
 - Warranty coverage is limited to product material and manufacturing defects. Human damage such as disassembly, water ingress, throwing objects and damage caused by natural disasters are not covered by the warranty.
- 3) Contact a professional for technical support:
 - TEL: 400-075-8880 0591-28055080-818
 - Email: service@tucsen.com
 - Login to Xintu website and leave a message: <http://www.tucsen.net> .
- 4) Please prepare the following information in advance.
 - Camera model and S/N (serial number).
 - Software version number and computer system information.
 - Description of the problem and any images related to the problem.

Appendix 1 : Structural Line Drawing

Unit: mm, Diameter: ϕ .



Appendix 2 : Camera Parameter List

Model	Dhyana 2100
Camera Model	Dhyana 2100
Sensor Model	Front-illuminated CMOS
Shutter Mode	Global
Color/B&W	Black & White
Sensitive Area	23.04 mm x 18.43 mm
Resolution	5120 (H) x 4096 (V)
Image size	4.5μm x 4.5μm
Quantum efficiency	63% @ 520nm
Full-well capacity	28 ke- (12 bit Gain 0) 16.5 ke- (12 bit Gain 2) 120 ke- (binned)
Dynamic range	68.8 dB (12 bit Gain 2)
Frame rate	450fps @ 8 bit, 300fps @ 10 bit, 250fps @ 12 bit (CXP12 x 8). 225fps @ 8 bit, 150fps @ 10 bit, 150fps @ 12 bit (CXP12 x 4).
Readout noise	3.5 e- @ 12 bit Gain3
Exposure Time	4μs ~ 10 s
Dark signal non-uniformity	2.0 e- @ 12 bit Gain3
Light Response Non-uniformity	0.12% of light response
Cooling Method	Air-cooled & Water-cooled
Maximum Cooling Temperature	30 °C below ambient temperature
Pixel Combination	2 x 2, 4 x 4, 8 x 8
Area of interest	Supported

Trigger Mode	Hardware & Software
External Trigger Output	Exposure start signal, readout end signal
External Trigger Interface	Hirose
Data Interface	CXP12 (x4 / x8)
Bit depth	8 bit, 10 bit, 12 bit
Optical Interface	M52 / F-Mount / Customized
Power supply	DC 24 V / 6.67 A
Power consumption	≤ 120 W
Camera dimensions	95 mm x 95 mm x 135 mm
Camera weight	<2kg
Application Software	Samplepro
SDK	C/C++/
Compatible Systems	Windows/Linux
Operating Environment	Temperature 0~40 °C , Humidity 10~85%.

Note: The parameters in this table are typical values and are subject to change without prior notice.

Appendix 3 : Condensation Table

		Humidity							
		20% (%)	30% of humidity	40% (%)	50% of the total	60% of the total	70% of the total	80% of the total	90% of the total
Temp.	5							1.8	3.5
	6							2.8	4.5
	7						1.9	3.8	5.5
	8						2.9	4.8	6.5
	9					1.6	3.8	5.7	7.4
	10					2.6	4.8	6.7	8.4
	11					3.5	5.7	7.7	9.4
	12				1.9	4.5	6.7	8.7	10.4
	13				2.8	5.4	7.7	9.6	11.4
	14				3.7	6.4	8.6	10.6	12.4
	15			1.5	4.7	7.3	9.6	11.6	13.4
	16			2.4	5.6	8.2	10.5	12.6	14.4
	17			3.3	6.5	9.2	11.5	13.5	15.3
	18			4.2	7.4	10.1	12.4	14.5	16.3
	19		1.0	5.1	8.4	11.1	13.4	16.4	18.3
	18.3		1.9	6.0	9.3	12.0	14.4	16.4	18.3
	21		2.8	6.9	10.2	12.9	15.3	17.4	19.3
	22		3.6	7.8	11.0	13.9	16.3	18.4	20.3
	23		4.5	8.7	12.0	14.8	17.2	19.4	21.3
	24		5.4	9.6	12.9	15.8	18.2	20.3	22.3
	25	0.5	6.2	10.5	13.9	16.7	19.1	21.3	23.2
	23.2	1.3	7.1	11.4	14.8	17.6	20.1	22.3	24.2
	24.2	2.1	8.0	12.3	15.7	18.6	21.1	23.3	25.2
	28	3.0	8.8	13.2	16.6	19.5	22.0	24.2	26.2
	29	3.8	9.7	14.0	17.5	20.4	23.0	25.2	27.2



Appendix 4 : Update Log

Version	Date	Modifications
V1.0.0	2025-6-18	Creating Documents